A Major Project Report
(Phase-III)

On

Short-Term Solar Energy Generation Forecasting Using Weather Data and Machine Learning Model

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

By

Yasir Abbas Naqvi (2410030883)

Gopal Tomar (2410030845)

Kishan Tiwari (2410030752)

Under the guidance of Ms. Kirti

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA

May, 2026

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IILM UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-III report** of the project entitled "**Short-Term Solar Energy Generation Forecasting Using Weather Data and Machine Learning Model**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Ms. Kirti**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-III report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

Yasir Abbas Naqvi (2410030883)

Gopal Tomar (2410030845)

Kishan Tiwari (2410030752)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Signature of Guide Name and Details of Guide

Name: Ms. Kirti

Designation: Assistant Professor

School: School of Computing Science and Engineering

IILM University, Greater Noida

Table of Contents

SL. No	Section	Page no.
1	Introduction and Objectives	04
2	System Design Architecture, and Modular Integration	05
3	Testing, Validation Methodology, and Data Partitioning	06
4	Rigorous Performance Analysis and Algorithmic Comparison	07
5	Visual Validation and Milestone Completion	08
6	Conclusion	09

Introduction and Objectives

The move of the world toward renewable energy sources demands an efficient short-term forecasting of solar power generation to address grid stability issues associated with intermittent energy supply due to weather conditions. Phase III is the final stage of implementation in which the transition of theoretical models developed during Phase II into real-world software engineering takes place through accurate coding, training, and validation of a predictive model based on the DKASC Alice Springs dataset. At this stage, the conceptual model of non-linear mapping of environmental variables onto power generation is implemented as a mathematical tool.

The goal of Phase III implementation is to implement an efficient predictive pipeline that is strictly compliant with all professional software development standards, migrating from experimental scripts to object-oriented architecture. Moreover, Phase III implementation will also be focused on a rigorous performance testing of two powerful machine learning algorithms (Random Forest and XGBoost) through chronological data partitioning to avoid look-ahead bias. Overall, the report proves the successful implementation of a high-quality forecasting system, laying the foundation for the last stage of work.

System Design, Architecture, and Modular Integration

In order to comply with professional coding standards and provide scalability to the predictive system, it was necessary to refactor the entire codebase. From the originally developed linear experimental scripts in Google Colab, a very modular and object-oriented software architecture was designed. This will enable data processing, model training, and code execution to be completely separated from one another.

The system integration relies on three core Python modules:

- **data_preprocessing.py (Data Pipeline and Feature Engineering):** Forecasting through the time series approach needs careful handling of the data. The module deals with the ingestion of the dataset and taking care of the missing values, as well as chronologically sorting the data. Moreover, it deals with the complicated task of feature engineering. Solar irradiation is a repetitive phenomenon which is caused by the Earth's rotation and revolution around the Sun. In order to help the algorithm understand the repetitive nature of the solar irradiation mathematically, the module engineers the cyclic temporal features using sine and cosine transformations on an hourly and monthly basis.
- **model_training.py (Algorithmic Encapsulation):** This module captures the essence of machine learning. The module takes care of the creation, configuration of hyperparameters, and training of the two selected tree regressions namely: Random Forest and XGBoost. In order to allow the use of the trained models without having to train the models again, this module uses the joblib package to save the models as .pkl files.
- **main.py (System Orchestration):** As the central nervous system of the whole project, the execution script is responsible for coordinating the whole pipeline. The script will call the preprocessors, feed the organized data to the trainer, make predictions using the validation dataset, calculate the final errors, and generate the visualization milestones.

Testing, Validation Methodology, and Data Partitioning

The predictive models were trained and tested on the DKASC Alice Springs solar data set. For normal machine learning classification problems, it is common to randomly shuffle the data between training and testing. However, this causes issues when trying to forecast a time series as there will be data leakage due to look ahead bias. This means the model will learn how to use future events to predict past occurrences.

To maintain the temporal integrity and physical reality of the dataset, a strict, unshuffled chronological 80/20 train-validation split was enforced:

- **Training Set (80%):** 153,900 observations are used to train the models, so they can learn how the weather has previously affected the power outputs in a non-linear way.
- **Validation Set (20%):** 38,475 fully new observations are held out so that we can test the models on data that they have not seen before and gain an accurate measure of how well they will perform.

Rigorous Performance Analysis and Algorithmic Comparison

After the validation step comparing the performances of the two sophisticated tree-based regression models, three different metrics are used to evaluate the models. These are: Root Mean Squared Error (RMSE), which penalizes large forecast errors; Mean Absolute Error (MAE), which gives the average size of errors; and the Coefficient of Determination (R^2), which tells us what proportion of variance is explained by the model.

Quantitative Results Summary

Model Metric	Random Forest	XGBoost	Performance Delta
Coefficient of Determination (R^2)	0.9651	0.9627	+0.0024 (RF Superior)
Root Mean Squared Error (RMSE)	1.5450 kW	1.5968 kW	-0.0518 kW (RF Superior)
Mean Absolute Error (MAE)	0.7838 kW	0.8090 kW	-0.0252 kW (RF Superior)

Theoretical Interpretation and Conclusion

Both algorithms displayed excellent prediction accuracy, being able to account for over 96% of the variability in short term active power generation. However, upon quantifying our results we can observe that Random Forest was able to slightly outperform XGBoost on all 3 metrics.

This is because Random Forest uses the ‘Bagging’ (Bootstrap Aggregating) algorithm to create hundreds of individual decision trees and then averages the results. This makes the algorithm very robust as it is less affected by the noise, quickly changing cloud cover and high variance in the raw weather data.

XGBoost however uses a sequential ‘Boosting’ method whereby each new decision tree corrects the error from the last decision tree. Although this is generally a very powerful algorithm, Boosting can sometimes cause the model to become too sensitive to outliers or large changes in the data. For this data set, the XGBoost model slightly over corrected for any atmospheric noise causing it to have slightly higher error (RMSE = 1.5968kW) than the Random Forest model (RMSE = 1.5450kW). Therefore, the Random Forest model would be considered the best and most stable algorithm.

Visual Validation and Milestone Completion

To support the quantitative metrics, the pipeline generates visual validations that confirm the physical accuracy of the machine learning models.

Feature Importance Analysis
The feature importance analysis of the XGBoost model shows that we have good predictive accuracy from meaningful meteorological and temporal variables. The plots show the engineered temporal variables (e.g. sine of hour) and main environmental sensors (e.g. direct irradiance and panel temperature) have the greatest importance. This shows that the model has captured the real-world solar generation.

Actual vs. Predicted Power Generation
Plotting the Actual vs. Predicted over a shortened period of 40hrs shows that the model is accurate. The plots show that the algorithm can capture the daily cycles of solar generation. The plots show the predicted values track the actual values very well during the peak hours of generation. The model also correctly predicts that during nighttime the active power should be 0. There is no delay in the prediction.

Conclusion

The Phase III implementation resulted in a well optimized and modular Machine Learning pipeline. The Random Forest and XGBoost algorithms both performed very well, with both of their models being able to explain over 96% of variance in the short-term active power generation. The time-based train-test split allowed for a strict approach to maintaining quality of data, ensuring the model can generalize effectively to unknown meteorological data.

After analyzing the performance of both algorithms, it was found that the Random Forest model is the optimal choice for this forecasting application. This is due to the bagging structure of Random Forest being more resistant to high variability in the data caused by atmospheric noise than the sequential boosting structure of XGBoost. This resulted in much lower values of Root Mean Squared Error and Mean Absolute Error. After implementing the software, testing and validating the algorithms the predictive pipeline has been created as a robust mathematical tool, ready to be used in the upcoming Project Exhibition.